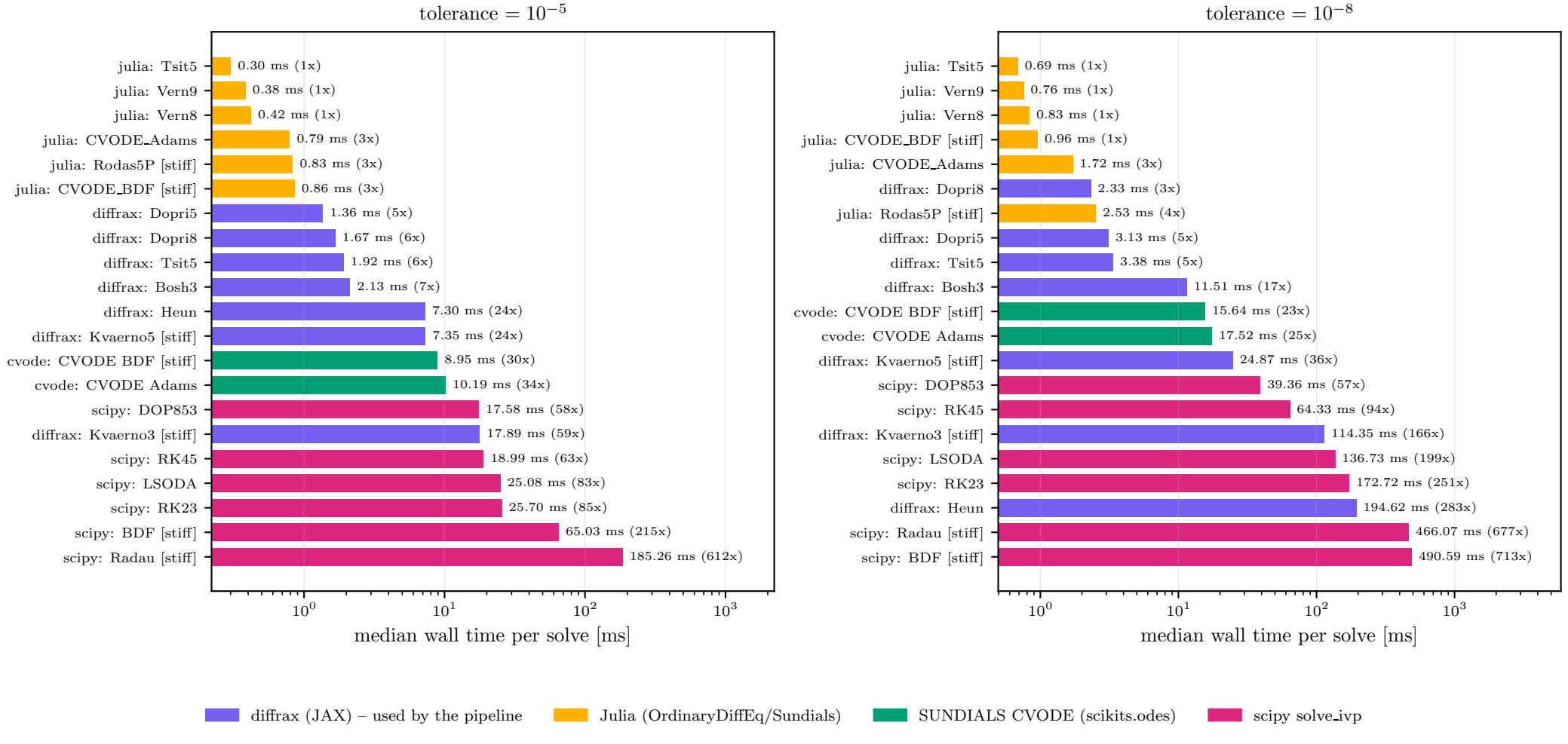
ODE solver speed on the TorC background equations (`_SAFE` point, identical problem and output grid)



Equal requested tolerance is NOT equal accuracy. These bars compare runtime at a requested `rtol`; solvers deliver wildly different *achieved* error for the same request (at 10^{-5}: scipy DOP853 reaches 8×10^{-7}, CVODE Adams only 5×10^{-3}). For the comparison that controls for this, see `plots/work_precision.pdf`.

Implicit solvers (Kvaerno) are configured with *explicit* root-finder tolerances — diffrax's default inherits them from the stepsize controller, which on this RHS costs a spurious 128×.

Single solves. In the pipeline the sampler batches: diffrax reaches ∼1.4 ms/solve under `vmap`, and the Julia IPC backend ∼0.19 ms/solve threaded — measured 3.0× end-to-end.